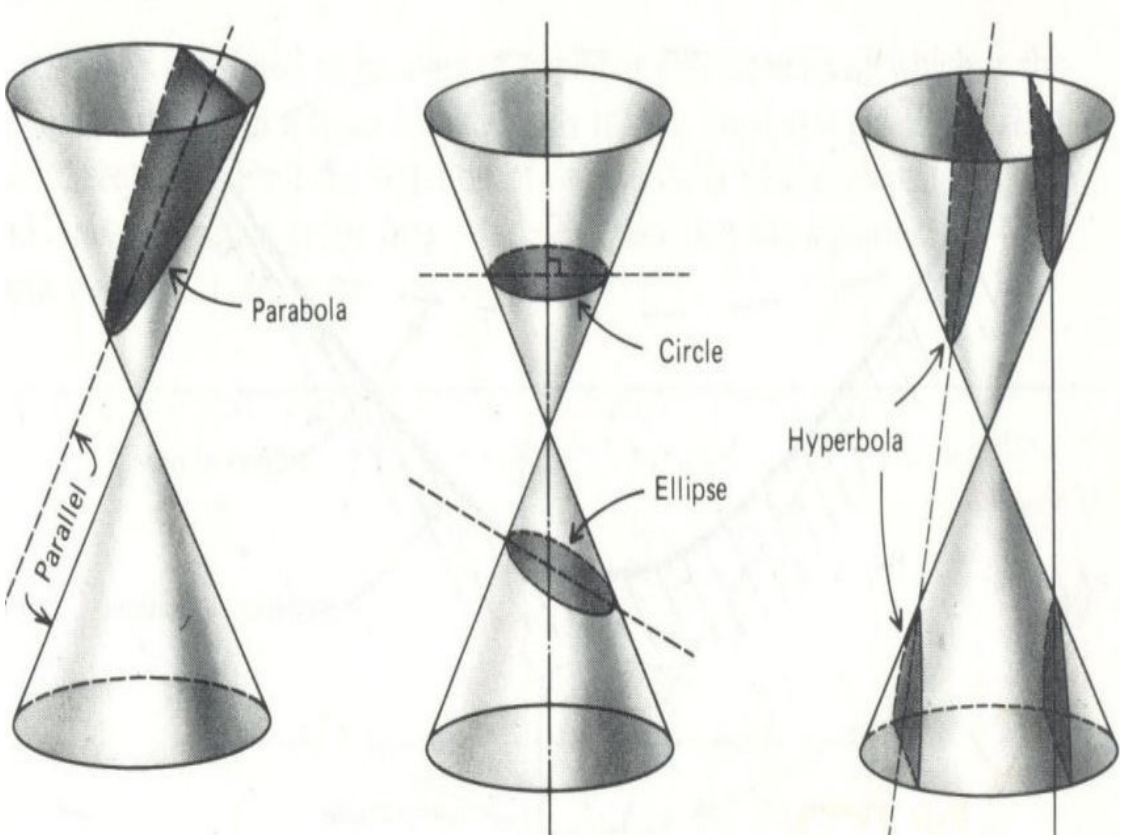
EMATH 100

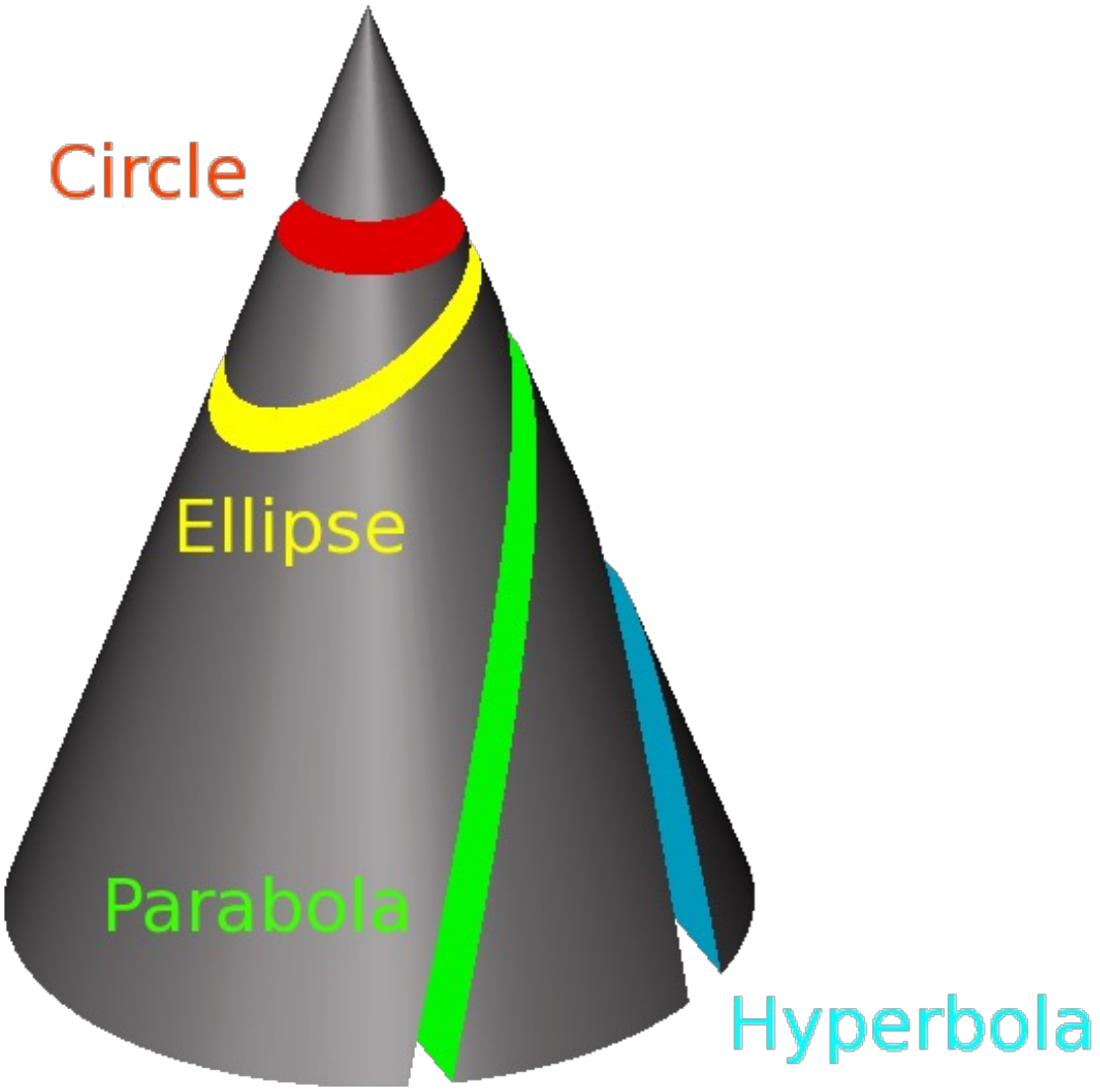
M. PAJA

Introduction to Conic Sections



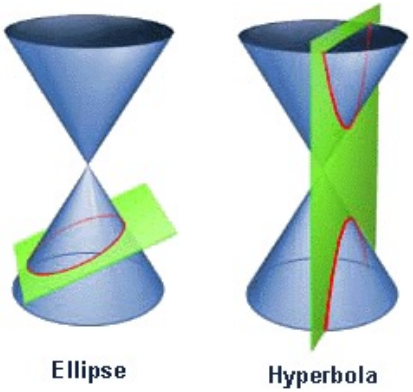
A conic section is a curve formed by the intersection of _____

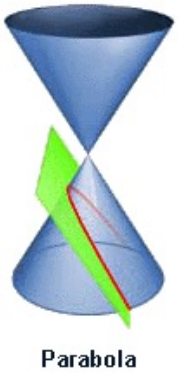
a plane and a double cone.


Circle
Ellipse
Parabola
Hyperbola

History

- Conic sections is one of the oldest math subject studied.
- The conics were discovered by Greek mathematician Menaechmus (c. 375-325 BC)
- Menaechmus's intelligence was highly regarded… he tutored Alexander the Great.



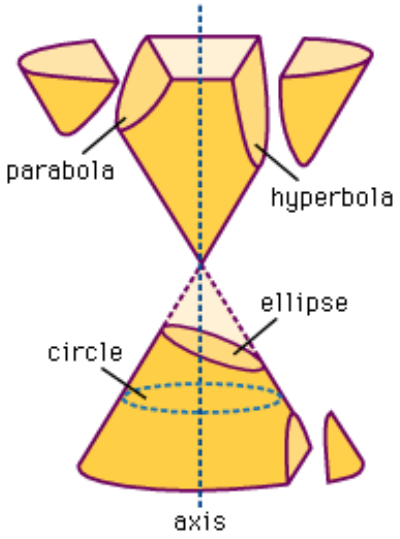


History

- Appollonius (c. 262-190 BC) wrote about conics in his series of books simply titled “Conic Sections”.
- Appollonious’ nickname was “the Great Geometer”
- He was the first to base the theory of all three conics on sections of one circular cone.
- He is also the one to give the name “ellipse”, “parabola”, and “hyperbola”.



Appollonius



Definitions

A. A **circle** is the set of all points in a plane that is a fixed distance from a fixed point

B. The **center** of a circle is the fixed equidistance point of the circle

C. The **radius** of a circle is the fixed equidistance length of the circle

D. A **tangent** is a line in the same plane as the circle that intersects at exactly one point

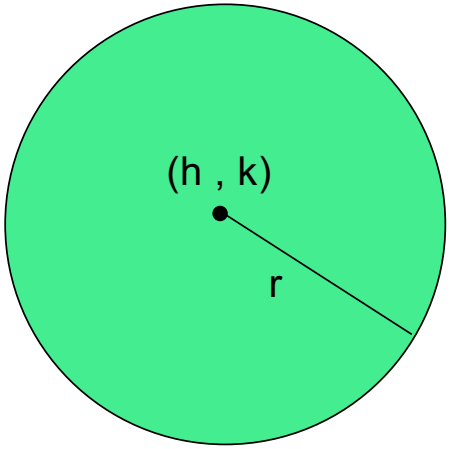
Circles

The set of all points that are the same distance from the center.

Standard Equation: $(x-h)^2+(y-k)^2=r^2$

CENTER: (h, k)

RADIUS: r (square root)



Example 1

Name the center and the radius of this circle, $(x + 2)^2 + (y - 3)^2 = 16$

$$(x-h)^2+(y-k)^2=r^2$$

$$r^2=16$$

$$r=4$$

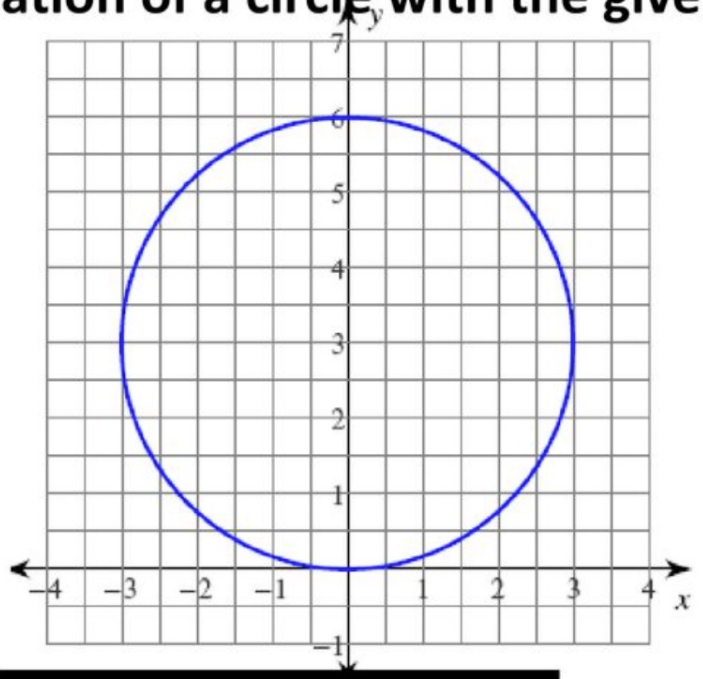
$$c=(-2,3)$$

$$r=4$$

Example 2

Identify the equation of a circle with the given graph below.

Each line is 0.5



$$x^2+(y-3)^2=9$$

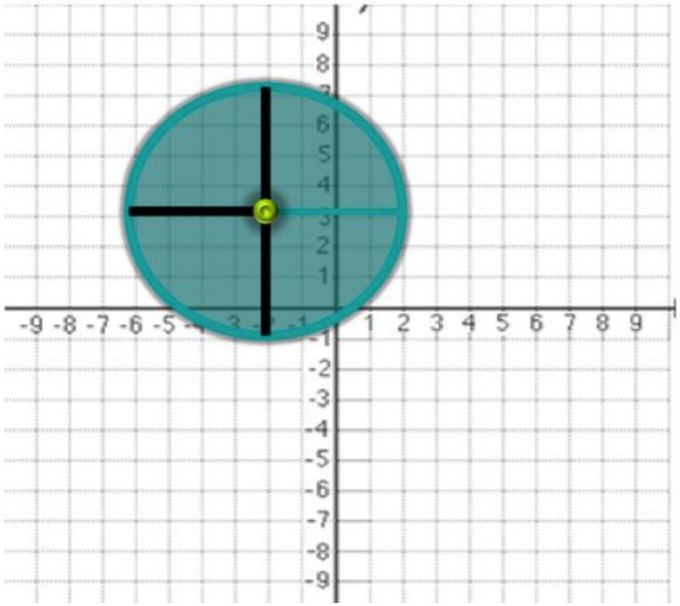
Example 3

Write an equation of the circle with its center is at (–2, 3), the radius is 4, and graph it.

$$(x-h)^2+(y-k)^2=r^2$$

$$(x+2)^2+(y-3)^2=4^2$$

$$(x+2)^2+(y-3)^2=16$$



Example 4

Write and graph the equation in standard form with the given center of (2, –3), the point (1, 0), and identify the radius.

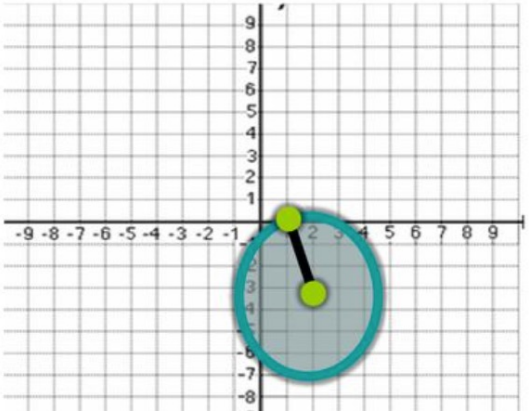
Plug in what you are given to figure out the radius

$$(x-2)^2+(y+3)^2=r^2$$

$$(1-2)^2+(0+3)^2=r^2$$

$$(-1)^2+(3)^2=r^2 \quad r=\sqrt{10}$$

$$\boxed{(x-2)^2+(y+3)^2=10}$$

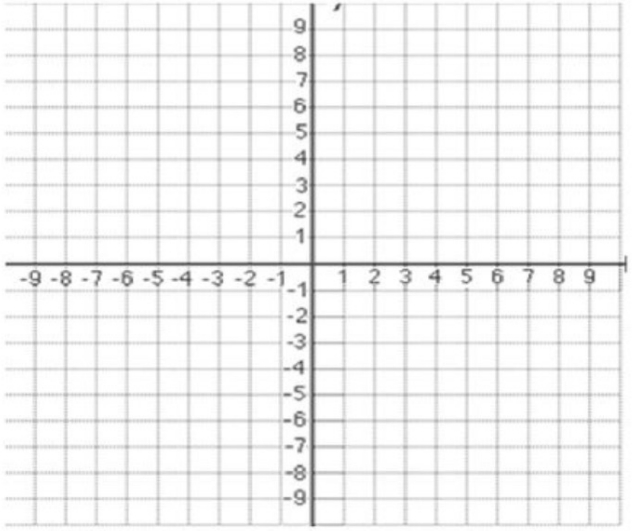


Example 5

Write a circle equation whose center is at the origin and passes through (1, –6).

$$(x-x_1)^2+(y-y_1)^2=r^2$$

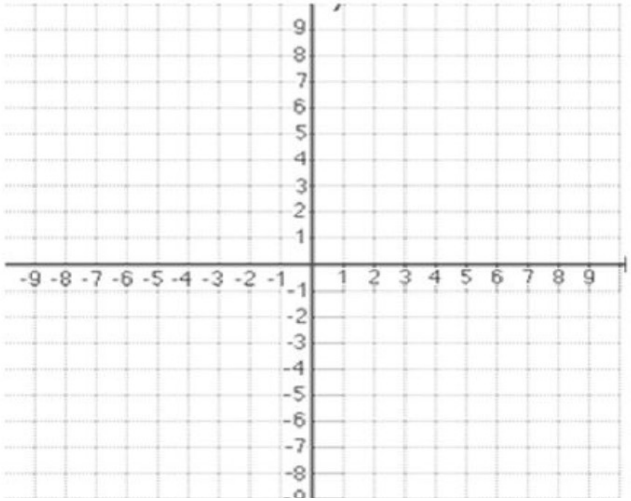
$$x^2+y^2=37$$



Your Turn

Write a circle equation whose endpoints of a diameter are (–5, 2) and (3, 6).

$$r = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} \qquad MP = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$



$$(x+1)^2 + (y-4)^2 = 20$$

Example 1

$$(x-2)^2+(y+8)^2=81$$

h

k

r²

Center: $(h,k) \rightarrow (2,-8)$

Radius: $r \longrightarrow 9$

Tangent Steps

A. **Determine the two points – usually the center and a point**
B. **Determine the slope of radius and take the <u>OPPOSITE SIGN RECIPROCAL</u>**
C. **Use point–slope formula to write equation**

Example 6

Write the equation of the line tangent to the circle $x^2 + y^2 = 29$ at the point (2, 5).

Put the equation in slope–intercept form using <u>POINT</u>–slope form with the center and point

$$y-(5)=-\frac{2}{5}(x-(2))$$

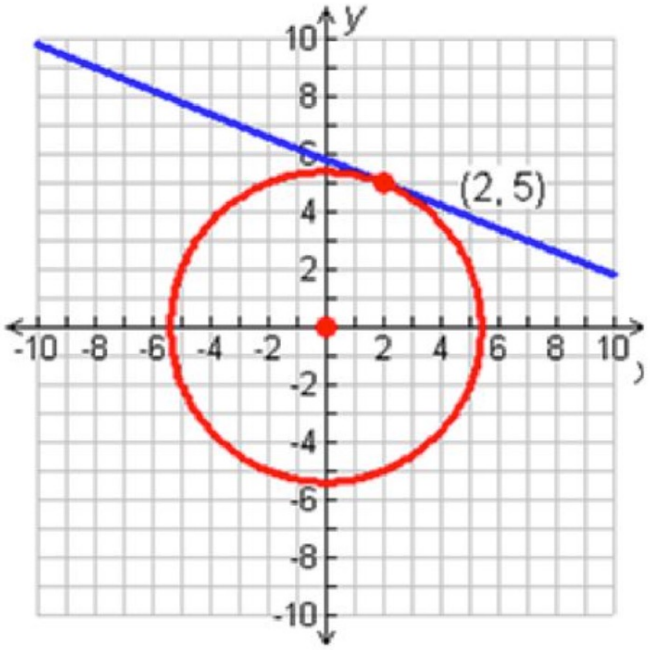
$$y-5=-\frac{2}{5}(x-2)$$

You can leave it in **POINT–SLOPE** form

Example 6

Write the equation of the line tangent to the circle $x^2 + y^2 = 29$ at the point (2, 5).

$$y - 5 = -\frac{2}{5}(x - 2)$$



Example 6

Write the equation of the line tangent to the circle $x^2 + y^2 = 29$ at the point (2, 5).

Put the equation in slope–intercept form using <u>POINT</u>–slope form with the center and point

$$m = \frac{y_2 - y_1}{x_2 - x_1} \qquad m = \frac{(5)-(0)}{(2)-(0)} \qquad m = \frac{5}{2}$$

The perpendicular lines are negative reciprocals, the slope of the tangent is –2/5.

$$m = -\frac{2}{5}$$

Write the equation of the line tangent to the circle $(x - 1)^2 + (y + 2)^2 = 25$ at the point (5, –5).

Write the equation of the line tangent to the circle $(x - 1)^2 + (y + 2)^2 = 26$ at the point on the circle (2, 3).

Given the equation of the circle,

$$x^2 + (y+1)^2 = 1$$

find the center and radius and graph it.

Write an equation of the circle with its center is at (–3, 5) and the radius is 5 and graph it.

$$y+5=\frac{4}{3}(x-5)$$

$$y-3=-\frac{1}{5}(x-2)$$

THE END.